

Global-guided Label-Aware Adaptive Multi-label Feature Selection via Fuzzy Mutual Information

Hao Xie, Yangding Li, Jianhua Dai*

Abstract—In recent years, embedded-based multi-label feature selection (MFS) has received increasing attention due to its ability to perform both feature selection and model training simultaneously. However, most existing methods tend to overlook the varying importance of different labels and typically rely on a weight matrix to implicitly conduct feature selection, which often leads to unstable results and feature redundancy. Moreover, these methods usually fail to fully exploit the global feature importance information inherent in the original data, thereby limiting further improvements in model performance. To address these issues, this paper proposes a novel global-guided label-aware adaptive MFS method based on fuzzy mutual information, called GLA-FMI. First, a convergent recursive propagation algorithm based on label correlations is designed to stably estimate label importance. Then, an adaptive feature activation strategy is designed, which explicitly selects features using binary activation vectors. Additionally, three complementary regularization terms are constructed to ensure that the model identifies an informative, compact, and interpretable subset of features. Specifically, the redundancy-aware regularization term prevents the simultaneous selection of highly redundant features, the feature sparsity regularization term enforces consistency between the binary activation vector and the weight matrix, and the global feature importance-based regularization term guides the model to prioritize more discriminative features. In comparative experiments, GLA-FMI ranks first in 77.33% of the 75 cases (15 datasets 5 evaluation metrics), significantly outperforming state-of-the-art MFS methods. The code is available at <https://github.com/xyy0717/GLA-FMI>.

Index Terms—Multi-label feature selection, fuzzy mutual information, label importance, adaptive feature activation, redundancy-aware.

I. INTRODUCTION

IN multi-label learning (MLL), each instance may be associated with multiple labels simultaneously [1]. Such problems are common in practical applications such as image annotation, text classification, sentiment analysis, and bioinformatics [2], [3]. As data dimensionality continues to increase, the original feature space often contains a large amount of redundant and irrelevant information, which not only increases the complexity of model training but may also degrade prediction performance and robustness [4], [5]. Therefore, selecting a subset of features that are most discriminative for multi-label tasks has become a critical challenge. Multi-label feature selection (MFS), as an effective dimensionality reduction technique, aims to identify the most relevant features with

respect to multiple labels, thereby improving the generalization ability, training efficiency, and interpretability of the model [6], [7].

MFS methods can be categorized into three types: filter-based [1], [8], wrapper-based [9], and embedded-based methods [10], [11]. Filter-based methods are known for their high efficiency, as they evaluate feature relevance based on pre-defined evaluation criteria without involving model training. Nevertheless, due to their independence from the learning model, they often struggle to deliver optimal predictive performance. Wrapper-based methods, on the other hand, rely on a specific learning model to evaluate different feature subsets, which generally leads to better accuracy but at the cost of increased computational complexity and limited scalability. Embedded-based methods distinguish themselves by incorporating feature selection directly into the model training process, typically through the optimization of a sparsity-inducing weight matrix W . This joint learning framework balances effectiveness with efficiency, making embedded-based methods increasingly popular in recent studies.

Although certain success has been achieved by existing embedded-based MFS methods, many limitations still exist. A notable issue is that most methods assume equal importance across all labels during the modeling process, overlooking the commonly observed imbalance in label importance in real-world multi-label tasks [12], [13]. In practical applications, some labels are significantly more important for overall task performance. For example, in medical diagnosis, identifying a critical condition such as cancer is often more important than detecting minor symptoms. Failing to consider such disparities may result in suboptimal feature selection, as the selected subset of features may neglect those critical for predicting high-priority labels. Therefore, incorporating label importance information into the feature selection process to enhance the model's focus on crucial labels remains a pressing challenge in MFS research.

Another significant limitation of existing embedded-based MFS methods stems from the sparsity-inducing regularization techniques commonly used to enhance the sparsity of the weight matrix W . Most methods adopt standard norms, such as the ℓ_1 -norm or $\ell_{2,1}$ -norm, to promote sparsity and highlight discriminative features [14], [15]. Although these techniques have demonstrated effectiveness in feature selection, they still suffer from two notable drawbacks. (1) Difficulty in obtaining a stable feature subset. These methods optimize to obtain a sparse weight matrix W and quantify the importance of the i -th feature using $\|W_i\|_2$. However, since the output is a continuous score rather than a discrete selection, the feature

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Hao Xie, Yangding Li and Jianhua Dai are with the Hunan Provincial Key Laboratory of Intelligent Computing and Language Information Processing, the College of Information Science and Engineering, Hunan Normal University, Changsha 410081, China (jhdai@hunnu.edu.cn).

subset becomes susceptible to noise and initialization, leading to instability. (2) Failure to address feature redundancy. Traditional sparsity norms focus solely on the individual relevance of features to the labels, without effectively considering the redundancy between features [16]. This may result in the simultaneous selection of highly redundant features, increasing model complexity and negatively affecting performance.

Moreover, existing embedded-based MFS methods typically rely solely on the weight matrix W to evaluate feature importance, overlooking the potential global importance information embedded in the original data [17], [18]. Due to the lack of a global assessment of overall feature relevance, the model can only depend on feedback during the training process to identify key features when optimizing W , making it difficult to promptly discover important features with potential value. Additionally, the absence of global guidance can make the model more susceptible to noise or local optima. Fuzzy mutual information, as an effective tool for uncovering latent information in data, has been widely applied in filter-based methods [19], [20]. Motivated by this, we attempt to explore global feature importance hidden in the raw data from the perspective of fuzzy mutual information to guide the feature selection process.

To address the above challenges, a *Global-guided Label-Aware Adaptive MFS* algorithm based on *Fuzzy Mutual Information*, named GLA-FMI, is proposed. First, a convergent recursive propagation algorithm based on label correlations is designed to estimate label importance and guide subsequent weighted learning. To enable explicit and controllable feature selection, an adaptive activation strategy with a binary activation vector s is proposed. Additionally, a redundancy-aware constraint regularization term is designed to suppress the simultaneous selection of highly redundant features, helping to reduce model complexity and improve generalization. Meanwhile, the feature sparsity regularization term enforces consistency between the binary activation vector s and the weight matrix W , further enhancing the model's interpretability. Finally, fuzzy mutual information is used to capture global feature importance from the original data, guiding the model to prioritize features with stronger label discriminability, thereby improving the overall performance of the model. The overall framework of GLA-FMI is illustrated in Figure 1. The main contributions of this paper are as follows:

- By modeling label correlations using fuzzy mutual information, a provably convergent recursive propagation algorithm is proposed to stably estimate label importance. It can then be used to guide subsequent weighted learning and enhance the model's focus on critical labels.
- Unlike most existing embedded MFS methods that implicitly perform feature selection through the weight matrix W , GLA-FMI designs an adaptive feature activation strategy that explicitly performs feature selection using a binary activation vector s .
- A redundancy-aware constraint and a feature sparsity regularization term are proposed to effectively prevent redundant features from coexisting and to ensure consistency between the activation vector s and the weight matrix W , thus enhancing the model's interpretability and

robustness.

- GLA-FMI guides the model to prioritize features with stronger label discriminability by extracting global feature importance from the raw data, thereby enhancing the discriminability and generalization performance of the selected feature subset.

The structure of this paper is as follows: Section II reviews related research. Section III introduces the necessary basic knowledge. Section IV describes the proposed model in detail. Section V presents a comparative analysis of the experimental results. Section VI summarizes the work of the paper.

II. RELATED WORK

In this section, related studies, including embedded-based methods and information theory-based methods, are briefly reviewed.

A. Embedded-based Methods

Embedded-based methods perform feature selection and learning simultaneously during model training, combining model optimization with feature selection. Sparse model-based methods are typical representatives, where sparse regularization (such as ℓ_1 and $\ell_{2,1}$ norms) is applied to model parameters, driving the weights of less important features towards zero for feature selection. The ℓ_1 -norm encourages column-wise sparsity, which helps identify features specific to individual labels. For instance, Zhang et al. [21] proposed the GLFS method, which simultaneously considers label group-specific features and label-specific features. Yang et al. [22] enhanced the learning of label-specific features by jointly considering both local and global label correlations within the ℓ_1 -norm framework. On the other hand, the $\ell_{2,1}$ -norm promotes row-wise sparsity, aiding in the selection of important features shared across multiple labels [23]. For instance, He et al. [24] proposed CLEMLFS method, which enhances label information through label distribution learning and latent sample correlation. Li et al. [25] proposed the SRFS method, which combines inner product regularization and $\ell_{2,1}$ -norm to address feature omission and redundancy. Wang et al. [26] proposed NMMFS method, which incorporates nonlinear mapping and manifold regularization to capture complex relationships between features and labels.

In addition, graph-based methods are a major branch of embedded-based methods. By incorporating graph regularization terms, they help preserve data structure information, improving feature selection performance and enhancing model robustness. For instance, Huang and Wu [27] proposed the MRDM method, which preserves the topological structure and enhances dependence with class labels by projecting data onto a low-dimensional manifold space. Zhang et al. [28] proposed LRDG method, which improves pseudo-label learning and minimizes information loss by utilizing latent representation learning and dynamic graph constraints. Gao et al. [29] proposed the SSFS method, which uses a latent structure shared term to preserve latent feature and label structures, and employs graph regularization to ensure consistency between the original feature space and the latent structure space.

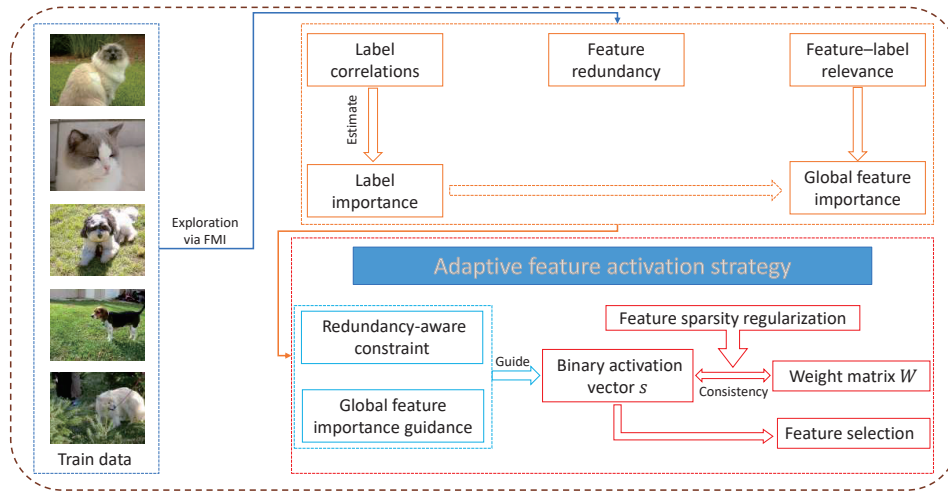


Fig. 1. The framework of the proposed GLA-FMI algorithm.

B. Information Theory-based Methods

Information theory-based methods typically evaluate feature importance by measuring the dependency between features and labels. Mutual information (MI), a commonly used metric, effectively captures such dependencies and has been widely applied in feature selection methods [30]. For instance, Zhang et al. [17] proposed the GRRO method, which integrates feature relevance, label correlations, and feature redundancy into a unified framework to achieve efficient MFS. the MDMR algorithm [31] utilizes MI to quantify both the relevance between features and labels and the redundancy among features, thereby constructing an evaluation function to guide the feature selection process. However, traditional mutual information is limited in handling continuous data. To address this, fuzzy mutual information (FMI) extends mutual information by incorporating fuzzy set theory, enabling a more flexible and accurate evaluation of feature-label dependencies in complex data scenarios [32]. In recent years, numerous MFS methods based on FMI have been proposed. For instance, Du et al. [33] proposed a fuzzy mutual information-based MFS method, which can handle streaming labels while capturing label dependencies. Dai et al. [4] proposed the FMIOR method, which uses FMI to avoid repeated entropy calculations and incorporates feature weights into an orthogonal regression optimization framework. Xiong et al. [34] proposed a method that combines label distribution learning with FMI to better capture label importance differences and enhance feature relevance evaluation.

In addition, Zentropy, as an extension of traditional entropy, introduces uncertainty measures, demonstrating superior stability and accuracy in handling complex and heterogeneous data [35]. Feature selection methods based on Zentropy have emerged and made significant progress. For instance, Yuan et al. [36] proposed an approach based on the neighborhood-aware fuzzy rough set and Zentropy measure, improving anti-noise ability and classification performance. Ye and Xu et al. [37] proposed a novel granular-balls generation method based on neighbor search, introducing multi-granularity granular-balls and combining them with generalized multi-granularity

rough sets and Zentropy for efficient feature selection. Yuan et al. [38] proposed a multigranularity data analysis approach with a Zentropy uncertainty measure for efficient and robust feature selection, integrating multigranularity information.

III. PRELIMINARIES

In this section, the basic knowledge and optimization techniques underlying the proposed method are introduced.

A. Fuzzy Similarity Relation

Definition 1. [39] Let $\mathcal{B}$ be a subset of real-valued attributes that describe a non-empty universe $\mathcal{S}$. The set $\mathcal{B}$ induces a fuzzy binary relation $\mathfrak{R}_{\mathcal{B}}$ on $\mathcal{S}$, which is called a fuzzy similarity relation if it satisfies reflexivity, i.e., $\mathfrak{R}_{\mathcal{B}}(x_i, x_i) = 1$, and symmetry, i.e., $\mathfrak{R}_{\mathcal{B}}(x_i, x_j) = \mathfrak{R}_{\mathcal{B}}(x_j, x_i)$ for any $x_i, x_j \in \mathcal{S}$.

Definition 2. [32], [39] Given a fuzzy similarity relation $\mathfrak{R}_{\mathcal{B}}$ on $\mathcal{S}$ induced by $\mathcal{B}$, its matrix form is defined as $MF(\mathfrak{R}_{\mathcal{B}}) = [r_{ij}]_{n \times n}$, where $r_{ij} = \mathfrak{R}_{\mathcal{B}}(x_i, x_j)$ denotes the fuzzy similarity between instances x_i and x_j .

The fuzzy similarity between two instances x_i and x_j can be calculated using the Gaussian kernel [40], [41]: $\mathfrak{R}_{\mathcal{B}}(x_i, x_j) = \bigcap_{b \in \mathcal{B}} \mathfrak{R}_b(x_i, x_j)$, where $\mathfrak{R}_b(x_i, x_j) = \exp\left(-\frac{\|x_i - x_j\|_b^2}{2\mu_b^2}\right)$, with $\|\cdot\|_b$ denoting the Euclidean distance and μ_b the standard deviation of b .

Definition 3. [42] Given a fuzzy similarity relation $\mathfrak{R}_{\mathcal{B}}$ on $\mathcal{S}$ induced by $\mathcal{B}$. The fuzzy similarity class of x_i induced by $\mathfrak{R}_{\mathcal{B}}$ is given by:

$$[s_i]_{\mathfrak{R}_{\mathcal{B}}} = \frac{r_{i1}}{x_1} + \frac{r_{i2}}{x_2} + \dots + \frac{r_{in}}{x_n}. \quad (1)$$

B. Information Measurement Method

Definition 4. [4] Given a fuzzy similarity relation $\mathfrak{R}_{\mathcal{B}}$ on $\mathcal{S}$ induced by $\mathcal{B}$. The fuzzy information entropy of $\mathcal{B}$ is defined as follows:

$$FH(\mathcal{B}) = -\frac{1}{n} \sum_{i=1}^n \log \frac{|[x_i]_{\mathcal{R}_{\mathcal{B}}}|}{n}, \quad (2)$$

where $|\cdot|$ represents the cardinality of the fuzzy similarity class.

Definition 5. [4] Given two fuzzy similarity relations $\mathfrak{R}_B$ and $\mathfrak{R}_C$ on $\mathcal{S}$ induced by B and C , respectively. The fuzzy mutual information between B and C is defined as follows:

$$FMI(B; C) = -\frac{1}{n} \sum_{i=1}^n \log \frac{|[x_i]_{\mathfrak{R}_B}| \times |[x_i]_{\mathfrak{R}_C}|}{n \times |[x_i]_{\mathfrak{R}_B} \cap [x_i]_{\mathfrak{R}_C}|}. \quad (3)$$

Definition 6. [4] Given two fuzzy similarity relations $\mathfrak{R}_B$ and $\mathfrak{R}_C$ on $\mathcal{S}$ induced by B and C , respectively. The normalized fuzzy mutual information between B and C is defined as follows:

$$NFMI(B; C) = \frac{2FMI(B; C)}{FH(B) + FH(C)}. \quad (4)$$

The normalized fuzzy mutual information $NFMI(B; C)$ quantifies the relative strength of association between B and C .

C. Iteratively Re-Weighted Method

The Iteratively Re-Weighted (IRW) method [43] is designed to solve the following optimization problem:

$$\min_{x \in \mathcal{U}} \mathcal{F}(x) + \sum_i \mathcal{H}_i(\mathcal{G}_i(x)), \quad (5)$$

where $\mathcal{U}$ denotes the feasible set for x , and $\mathcal{H}_i(\mathcal{G}_i(x))$ is a concave function w.r.t. $\mathcal{G}_i(x)$. Each $\mathcal{G}_i(x)$ may represent a scalar, vector, or matrix.

By introducing an intermediate variable $\mathcal{D}_i = \mathcal{H}'_i(\mathcal{G}_i(x))$, which represents the super(sub)-gradient of $\mathcal{H}_i$ at $\mathcal{G}_i(x)$, the original problem (5) can be equivalently transformed into:

$$\min_{x \in \mathcal{U}} \mathcal{F}(x) + \sum_i \text{Tr}(\mathcal{D}_i^\top \mathcal{G}_i(x)). \quad (6)$$

If the problem in (5) is a maximization problem, and $\mathcal{H}_i(\mathcal{G}_i(x))$ is convex w.r.t. $\mathcal{G}_i(x)$, then the optimization problem can be similarly transformed into an expression analogous to (6), with the objective function changed from minimization to maximization.

IV. THE PROPOSED MODEL

In this section we will introduce in detail the MFS method GLA-FMI proposed in this paper.

A. Problem Formulation and Notations

Given a feature matrix $X \in \mathbb{R}^{n \times m}$ and a label matrix $Y \in \{0, 1\}^{n \times l}$, where n , m , and l denote the number of instances, features, and labels, respectively, each entry Y_{ij} indicates whether the i -th instance is associated with the j -th label ($Y_{ij} = 1$) or not ($Y_{ij} = 0$). MFS aims to identify a subset of informative features that are most relevant to the labels.

To facilitate notation, we use $X_{i\cdot}$, $X_{\cdot j}$, and X_{ij} to represent the i -th row, j -th column, and the element at position (i, j) of X , respectively. Additionally, commonly used matrix operations include the transpose (X^T), the trace ($\text{tr}(X)$), and the Frobenius norm ($\|X\|_F$). Some key symbols used throughout this paper are summarized in Table I.

TABLE I
SUMMARY OF KEY SYMBOLS.

Symbol	Description
$X \in \mathbb{R}^{n \times m}$	Feature matrix.
$Y \in \{0, 1\}^{n \times l}$	Label matrix.
n	Number of instances.
m	Number of features.
l	Number of labels.
$\mathcal{W} \in \mathbb{R}^{l \times l}$	Label correlation matrix.
$A \in \mathbb{R}^{l \times l}$	Label connection matrix.
$\theta \in \mathbb{R}^l$	Label importance vector.
σ	Damping factor.
$v \in \mathbb{R}^l$	Normalized initial vector.
$W \in \mathbb{R}^{m \times l}$	Weight matrix.
$s \in \{0, 1\}^{m \times 1}$	Binary activation vector.
$\hat{R} \in \mathbb{R}^{m \times m}$	Feature redundancy matrix.
$U \in \mathbb{R}^{m \times l}$	Feature-label relationship matrix.
$g = U\theta \in \mathbb{R}^m$	Global feature importance vector.
$\Theta = \text{diag}(\theta) \in \mathbb{R}^{l \times l}$	Diagonal matrix constructed from θ .
α, β, γ	Balancing parameter.
r_1, r_2, r	Penalty coefficient.
k	Number of features to select.

B. Label Importance Estimation

In MLL, not all labels contribute equally to the learning process. It is crucial to estimate the importance of each label for more effective MFS. In this section, a recursive propagation algorithm based on label correlations is designed to estimate label importance effectively.

To capture the interdependencies among labels, we construct a weighted undirected graph $\mathcal{G} = (\mathcal{L}, \mathcal{E}, \mathcal{W})$, where each node $l_i \in \mathcal{L}$ represents the i -th label, each edge $(l_i, l_j) \in \mathcal{E}$ indicates that there exists a correlation between labels i and j , and $\mathcal{W}_{ij}$ denotes the strength of this correlation, computed as $\mathcal{W}_{ij} = NFMI(l_i; l_j)$. In this graph, a label is considered more important if it is strongly connected to other important labels. This mutual reinforcement is naturally modeled through a recursive propagation mechanism, as described below:

$$\theta^{t+1} = \sigma A \theta^t + (1 - \sigma)v, \quad (7)$$

where θ^t denotes the label importance vector at the t -th iteration, σ is a damping factor (typically set to 0.85), and v is a normalized initial vector, usually defined as $v = \frac{1}{l} \mathbf{1}^{l \times 1}$. The matrix $A \in \mathbb{R}^{l \times l}$ is a column-normalized label connection matrix, where each entry A_{ij} is defined as

$$A_{ij} = \begin{cases} \frac{e^{\mathcal{W}_{ij}} - 1}{\sum_{k \neq j} (e^{\mathcal{W}_{kj}} - 1)} & \text{if } i \neq j, \\ 0 & \text{otherwise.} \end{cases} \quad (8)$$

It encodes the relative influence of the i -th label on the j -th label based on their correlation strength. The exponential transformation amplifies strong correlations and suppresses weak ones, while the removal of diagonal elements ensures that the importance of a label is determined solely by its relationships with other labels.

By recursively expanding Eq. (7), the label importance at iteration $t + 1$ can be expressed as follows:

$$\theta^{t+1} = \sigma^t A^t \theta^1 + (1 - \sigma) \sum_{i=0}^{t-1} \sigma^i A^i v. \quad (9)$$

Since $0 < \sigma < 1$ and A is a column-normalized matrix, the sequence $\{\theta^t\}$ forms a convergent series (the convergence analysis will be presented in the next subsection). As $t \rightarrow \infty$, the recursive process stabilizes, and $\{\theta^t\}$ converges to a unique fixed point:

$$\theta^* = (1 - \sigma)(I - \sigma A)^{-1}v. \quad (10)$$

Obviously, $\sum_i \theta_i^* = 1$, and it is worth noting that the convergence result is independent of the initial state θ^1 , ensuring the stability and consistency of the label importance estimation.

Example 1. Given a non-empty universe $\mathcal{S} = \{x_1, x_2, x_3, x_4\}$ and a label set $L = \{l_1, l_2, l_3\}$. The detailed labeling information is shown in Table II. The procedure for calculating the importance of each label is as follows:

First, based on Definition 2, the fuzzy similarity relation induced by each label can be computed individually. Then, by using Eqs. (4) and (8), two key relation matrices, $\mathcal{W}$ and A , can be obtained, as shown below:

$$\mathcal{W} = \begin{bmatrix} 1 & 0.49 & 0.28 \\ 0.49 & 1 & 0.49 \\ 0.28 & 0.49 & 1 \end{bmatrix}, A = \begin{bmatrix} 0 & 0.50 & 0.34 \\ 0.66 & 0 & 0.66 \\ 0.34 & 0.50 & 0 \end{bmatrix}.$$

Finally, based on Eq. (10), we can obtain the label importance vector $\theta = [0.3042, 0.3916, 0.3042]$.

TABLE II
A SIMPLE EXAMPLE.

	l_1	l_2	l_3
x_1	1	0	1
x_2	0	0	1
x_3	0	1	0
x_4	0	1	1

C. Convergence Analysis of the Sequence $\{\theta^t\}$

In this subsection, the convergence of the sequence $\{\theta^t\}$ defined in Eq. (9) is analyzed, with the goal of showing that it converges to a unique fixed point as $t \rightarrow \infty$. To this end, the following two lemmas are introduced [44].

Lemma 1. For $\mathcal{Q} \in \mathbb{R}^{n \times n}$, $\lim_{t \rightarrow \infty} \mathcal{Q}^t = 0$ if and only if its spectral radius $\rho(\mathcal{Q}) < 1$.

Lemma 2. If $\|\cdot\|$ is a matrix norm and $\|\mathcal{Q}\| < 1$, then $I_n - \mathcal{Q}$ is nonsingular and $(I_n - \mathcal{Q})^{-1} = \sum_{t=1}^{\infty} \mathcal{Q}^t$.

Since A is a column-normalized matrix, its spectral radius satisfies $\rho(A) \leq 1$. Additionally, since $0 < \sigma < 1$, the following holds:

$$\begin{cases} \rho(\sigma A) < 1, \\ \|\sigma A\| < 1. \end{cases} \quad (11)$$

Applying Lemmas 1 and 2, we obtain:

$$\begin{cases} \lim_{t \rightarrow \infty} (\sigma A)^t = 0, \\ \sum_{t=1}^{\infty} (\sigma A)^t = (I_n - \sigma A)^{-1}. \end{cases} \quad (12)$$

Thus, the sequence $\{\theta^t\}$ converges, and the limit is given by:

$$\theta^* = \lim_{t \rightarrow \infty} \theta^{t+1} = (1 - \sigma)(I - \sigma A)^{-1}v. \quad (13)$$

D. Method Framework

In MLL, recognizing the varying significance of labels can facilitate the capture of relationships between features and labels. To this end, this paper proposes a weighted loss function that accounts for label importance, aiming to learn a weight matrix W that maps the feature matrix X to the label matrix Y while minimizing the weighted loss. The weighted loss function is defined as follows:

$$\min_W \sum_{i=1}^n \sum_{j=1}^l \theta_j (XW - Y)_{ij}, \quad (14)$$

which can be rewritten as:

$$\min_W \|(XW - Y)\Theta^{\frac{1}{2}}\|_F^2, \quad (15)$$

where Θ is a diagonal matrix constructed from the label importance vector θ , defined as $\Theta = \text{diag}(\theta_1, \theta_2, \dots, \theta_l)$. The values of θ can be computed using Eq.(10).

To achieve discrete and interpretable feature selection, this paper proposes an adaptive feature activation strategy based on a binary activation vector $s \in \{0, 1\}^{m \times 1}$, where $s_i = 1$ indicates that the i -th feature is activated (i.e., selected), and $s_i = 0$ indicates that the i -th feature is unactivated (i.e., not selected). First, to effectively prevent redundant features from impairing model performance, a redundancy-aware regularization term is introduced to penalize the simultaneous selection of highly redundant feature pairs. This idea can be formalized as:

$$\min_s \alpha s^T \hat{R} s, \quad (16)$$

where α is a balancing parameter, and $\hat{R}$ is a feature redundancy matrix, whose diagonal elements are zero (i.e., $\hat{R}_{ii} = 0$), and the off-diagonal elements are defined as $\hat{R}_{ij} = NFM I(f_i; f_j)$ for $i \neq j$, where f_i and f_j represent the i -th and j -th features, respectively.

Next, to ensure the consistency between the weight matrix W and the binary activation vector s , a new feature sparsity regularization term is introduced. This regularization term penalizes the expression of unactivated features, making the model more reliant on the activated feature subset, thereby enhancing the model's interpretability.

$$\min_{W,s} \beta \sum_{i=1}^m (1 - s_i) \|W_i\|_2^{r_1}, \quad (17)$$

where β is a balance parameter, r_1 is a penalty coefficient, and $\|W_i\|_2$ denotes the ℓ_2 -norm of the i -th row of W , which reflects the overall strength of the i -th features contribution to all labels.

Finally, to further enhance the reliability of the activated features, a global feature importance-based regularization term is proposed. Specifically, by quantifying and weighting the importance of each feature, the optimization process is guided

to prioritize the activation of more discriminative features, thereby improving the discriminability and stability of the selected feature subset.

$$\min_s \gamma \sum_{i=1}^m (1 - s_i) g_i^{r_2}, \quad (18)$$

where γ is a balancing parameter, r_2 is a penalty coefficient, and $g = U\theta$ is a global feature importance vector. Here, g_i represents the global importance of the i -th feature, with U as the feature-label relationship matrix and $U_{ij} = NMFI(f_i; l_j)$.

By combining Eqs. (15), (16), (17), and (18), the final objective function is obtained, with the constraint $s^\top s = k$ ensuring the selection of exactly k features.

$$\begin{aligned} \min \mathcal{L}(W, s) = & \|(XW - Y)\Theta^{\frac{1}{2}}\|_F^2 + \alpha s^\top \hat{R}s \\ & + \beta \sum_{i=1}^m (1 - s_i) \|W_i\|_2^{r_1} + \gamma \sum_{i=1}^m (1 - s_i) g_i^{r_2}, \quad (19) \\ \text{s. t. } & s \in \{0, 1\}^{m \times 1}, s^\top s = k. \end{aligned}$$

It is worth noting that GLA-FMI has the following advantages: Unlike many existing MFS methods that implicitly perform feature selection by relying on the weight matrix W , GLA-FMI proposes an adaptive feature activation strategy based on the binary vector s for explicit feature selection. This mechanism enhances the interpretability of the model and improves control over the selection process. By using s , we can directly identify which features are activated and which are not, making the model more transparent and focused on the most informative features. Moreover, by designing redundancy-aware, feature sparsity regularization, and global feature importance-based regularization constraints into the objective function, GLA-FMI ensures that the selected feature subset is both informative and compact, thereby enhancing the model's generalization ability and overall performance.

E. Solution

In this section, the optimization problem in Eq. (19) is solved using an alternating iteration method. In this method, individual variables are updated sequentially while other variables remain fixed.

1) *Update W*: With s fixed, the update of W reduces the original problem in (19) to the following simplified form:

$$\mathcal{L}(W) = \|(XW - Y)\Theta^{\frac{1}{2}}\|_F^2 + \beta \sum_{i=1}^m (1 - s_i) \|W_i\|_2^{r_1}. \quad (20)$$

For convenience, we introduce a diagonal matrix D to rewrite $\sum_{i=1}^m (1 - s_i) \|W_i\|_2^{r_1}$ as $\text{tr}(W^T D W)$, where $D_{ii} = \frac{r_1(1-s_i)}{2\|W_i\|_2^{2-r_1}}$. This results in the following reformulated problem:

$$\hat{\mathcal{L}}(W) = \|(XW - Y)\Theta^{\frac{1}{2}}\|_F^2 + \beta \text{tr}(W^T D W). \quad (21)$$

The gradient of $\hat{\mathcal{L}}(W)$ w.r.t. W is given by:

$$\frac{\partial \hat{\mathcal{L}}(W)}{\partial W} = 2X^T(XW\Theta - Y\Theta) + 2\beta DW. \quad (22)$$

Setting the gradient in Eq. (22) to zero gives an update rule for W , which is difficult to solve directly. Therefore, we adopt

a column-wise updating strategy. Specifically, by isolating the j -th column from Eq. (22) and setting it to zero, we obtain:

$$\theta_j X^T(XW_{\cdot j} - Y_{\cdot j}) + \beta DW_{\cdot j} = 0. \quad (23)$$

Solving Eq. (23) yields the update rule for the j -th column of W :

$$W_{\cdot j} = (X^T X \theta_j + \beta D)^{-1} X^T Y_{\cdot j} \theta_j. \quad (24)$$

2) *Update s*: With W fixed, the update of s reduces the original problem in Eq. (19) to the following simplified form by removing constant terms irrelevant to s :

$$\min_s \alpha s^\top \tilde{R}s - s^\top z, \quad \text{s. t. } s \in \{0, 1\}^{m \times 1}, s^\top s = k, \quad (25)$$

where $z \in \mathbb{R}^m$ is a vector with entries defined as $z_i = \beta \|W_i\|_2^{r_1} + \gamma g_i^{r_2}$. Solving Eq. (25) is challenging due to its non-convexity, and thus we apply the following relaxation:

$$\max_{s \in \{0, 1\}^{m \times 1}, s^\top s = k} \alpha s^\top \tilde{R}s + s^\top z, \quad (26)$$

where $\tilde{R} = cI_m - \hat{R} \in \mathbb{R}^{m \times m}$, and c is a relaxation parameter chosen to ensure that $\tilde{R}$ is a positive semi-definite matrix. For convenience, we set the largest eigenvalue of $\hat{R}$, denoted as $\lambda_{\max}(\hat{R})$, to be c , i.e., $c = \lambda_{\max}(\hat{R})$.

Clearly, Eq. (26) is a convex function w.r.t. s . Thus, it can be solved using the IRW method. Specifically, s is updated by solving the following problem in each iteration:

$$\max_{s \in \{0, 1\}^{m \times 1}, s^\top s = k} s^\top f, \quad (27)$$

where $f = 2\alpha \tilde{R}s + z \in \mathbb{R}^{m \times 1}$, the gradient of the objective function in Eq. (26) w.r.t. s .

To update s , the elements of f are sorted in descending order, and the top k indices are selected. The corresponding positions in s are set to 1, while the remaining positions are set to 0. Formally, the update rule can be written as:

$$s_i = \begin{cases} 1, & \text{if } i \in \mathbb{I}_{\text{top-}k}(f), \\ 0, & \text{otherwise,} \end{cases} \quad (28)$$

where $\mathbb{I}_{\text{top-}k}(f)$ returns the indices of the top k elements in f .

According to the above optimization results, we can give the detailed algorithm of GLA-FMI, as shown in Algorithm 1.

F. Convergence Analysis of Algorithm 1

To demonstrate that the iterative updates of variables W and s in Algorithm 1 effectively reduce the objective function and guarantee convergence, we introduce a lemma from [45].

Lemma 3. For $0 < p \leq 2$, the following inequality holds for any nonzero vectors x_1 and x_2 :

$$\|x_1\|_2^p - \frac{p}{2} \frac{\|x_1\|_2^2}{\|x_2\|_2^{2-p}} \leq \|x_2\|_2^p - \frac{p}{2} \frac{\|x_2\|_2^2}{\|x_2\|_2^{2-p}}. \quad (29)$$

Proposition 1. The optimization procedure in Algorithm 1 progressively reduces the value of the objective function (19) until convergence is reached.

Algorithm 1 The GLA-FMI approach.

Require: The feature matrix X , the label matrix Y , and the parameters $\alpha, \beta, \gamma, r_1$, and r_2 , and the number of selected features k .

Ensure: Binary activation vector s .

- 1: Compute the relationship matrices $\mathcal{W}$, $\hat{R}$, and U based on the formula given in Eq. (4).
- 2: Calculate A using Eq. (8), then calculate θ using Eq. (10), and initialize $g = U\theta$.
- 3: Randomly initialize the weight matrix W , the binary vector s , and set the number of initial iterations to $T = 0$.
- 4: **repeat**
- 5: Compute the diagonal matrix D from W .
- 6: Update W column-wise by Eq. (24).
- 7: Update s by Eq. (27).
- 8: $T = T + 1$.
- 9: **until** convergence
- 10: **return** Binary activation vector s .

Proof. For simplicity, let $\tilde{W}$ and $\tilde{s}$ denote the updated results of W and s at each iteration, respectively. First, we prove that updating W does not increase the value of objective function (19).

Since Eq. (21) defines a convex function w.r.t. W , the updated solution $\tilde{W}$ is the optimal solution to the subproblem (21), which implies that:

$$\hat{\mathcal{L}}(\tilde{W}) \leq \hat{\mathcal{L}}(W). \quad (30)$$

According to Lemma 3, the following inequality holds:

$$\sum_i ((1 - s_i) \|\tilde{W}_i\|_2^{r_1} - \frac{r_1(1 - s_i)}{2} \frac{\|\tilde{W}_i\|_2^2}{\|\tilde{W}_i\|_2^{2-r_1}}) \leq \sum_i ((1 - s_i) \|W_i\|_2^{r_1} - \frac{r_1(1 - s_i)}{2} \frac{\|W_i\|_2^2}{\|W_i\|_2^{2-r_1}}). \quad (31)$$

By combining Eqs. (30) and (31), we obtain:

$$\mathcal{L}(\tilde{W}, s) \leq \mathcal{L}(W, s). \quad (32)$$

Next, since the updated $\tilde{s}$ is the optimal solution of problem (25), the following inequality holds:

$$\mathcal{L}(W, \tilde{s}) \leq \mathcal{L}(W, s). \quad (33)$$

Combining Eqs. (32) and (33), we can get the following inequality:

$$\mathcal{L}(\tilde{W}, \tilde{s}) \leq \mathcal{L}(W, s). \quad (34)$$

Therefore, Proposition 1 is proved.

G. Computational Complexity Analysis

The overall computational process of GLA-FMI is divided into two stages: the initialization stage and the optimization iteration stage. In the first stage, the task is to compute the label importance vector θ , which includes the construction of the relationship matrix $\mathcal{W}$ and the computation of matrix A . These operations involve matrix multiplication and inversion, with a time complexity of $O(n^2l^2 + l^3)$. In addition, it is

necessary to initialize the relationship matrix $\hat{R}$, matrix U , and vector g , with time complexities of $O(n^2m^2)$, $O(n^2ml)$, and $O(ml)$, respectively. In the second stage, the task is the alternating update of variables W and s . The update of the weight matrix W involves matrix addition, multiplication, and inversion, with a time complexity of $O(nm^2 + nml + m^3l)$. The update of the binary activation vector s involves matrix addition, multiplication, dominant eigenvalue computation, and matrix inversion, with a time complexity of $O(m^2)$. In summary, the total time complexity of GLA-FMI is $O(n^2l^2 + l^3 + n^2m^2 + n^2ml + T(nm^2 + nml + m^3l))$, where T denotes the number of iterations in the optimization stage.

TABLE III
DATASET DETAILS.

Dataset	Instances	Train	Test	Features	Labels
Arts	5000	2000	3000	462	26
Business	5000	2000	3000	438	30
Computers	5000	2000	3000	681	33
Entertainment	5000	2000	3000	640	21
Health	5000	2000	3000	612	32
Image	2000	1000	1000	294	5
Medical	978	333	645	1449	45
PlantGo	978	588	390	3091	12
Recreation	5000	2000	3000	606	22
Reference	5000	2000	3000	793	33
Scene	2407	1211	1196	294	6
Slashdot	3782	1891	1891	1079	22
Social	5000	2000	3000	1047	39
Society	5000	2000	3000	636	27
Yeast	2417	1500	917	103	14

V. EXPERIMENTS

In this section, extensive performance comparison experiments were conducted on fifteen multi-label datasets, with the proposed algorithm being evaluated against nine state-of-the-art MFS algorithms.

A. Experiment Preparation

1) *Datasets and Classifier Settings:* Experiments were conducted on fifteen widely used multi-label datasets^{1 2 3}, with detailed dataset characteristics summarized in Table III. All evaluations were performed using the ML-KNN classifier, where the number of neighbors was fixed at $K = 10$ to ensure a uniform comparison across all methods.

2) *Evaluation Metrics:* In this section, we introduced five evaluation metrics that were utilized in our experiments: Coverage (CV), Average Precision (AP), One Error (OE), Hamming Loss (HL), and Ranking Loss (RL). These metrics were widely adopted to evaluate the quality of predicted label rankings in multi-label learning tasks [46]. The test set was denoted as $\mathcal{D}_S = \{(x_i, \mathcal{Y}_i) \mid 1 \leq i \leq n_1\}$, where $\mathcal{Y}_i$ represented the ground-truth label set associated with instance x_i . The predicted label set was denoted by $\hat{\mathcal{Y}}_i$, and $h(x_i, l_j)$ indicated the real-valued output of the classifier for instance x_i on label l_j . A ranking function $\text{rank}(x_i, l_j)$ was defined such

¹<http://mulan.sourceforge.net/documentation.html>

²<http://www.uco.es/kdis/mlresources/>

³<https://sci2s.ugr.es/keel/multilabel.php>

TABLE IV
EXPERIMENTAL RESULTS FOR THE COMPARISON OF CV ($\downarrow$).

	SRFS	MRDM	GRRO	CLEMLFS	SSFS	GLFS	LRDG	FMIOR	NMMFS	GLA-FMI
Arts	5.1573	5.2593	5.6657	5.2293	5.4497	5.2067	5.3120	5.3050	5.3150	5.1723
Business	2.2560	2.2593	2.4020	2.2993	2.3313	2.2217	2.3090	2.3060	2.2837	2.2503
Computers	4.2537	4.3447	4.5870	4.3923	4.4813	4.2997	4.2937	4.3110	4.4620	4.2940
Entertainment	3.1613	3.1393	3.3737	3.1590	3.2827	3.1670	3.1430	3.1983	3.1997	3.0170
Health	3.0200	3.3317	3.7127	3.2133	3.3710	3.1397	3.1663	3.2713	3.2070	2.9907
Image	1.1760	3.2413	1.1140	3.1097	1.1510	3.1813	1.1510	1.1720	1.1200	1.0330
Medical	3.3674	2.8682	6.0667	2.6264	3.3597	2.8868	3.7984	4.8775	3.4078	2.5829
PlantGo	0.6667	0.6128	0.6590	0.4949	0.6692	0.5103	1.5949	1.0051	0.6385	0.4872
Recreation	4.4870	4.7150	5.3190	4.5703	4.9690	4.6113	4.8177	4.7183	4.8140	4.4460
Reference	3.1750	3.3753	3.6833	3.2613	3.3500	3.2480	3.3507	3.2283	3.1857	3.0910
Scene	0.6446	0.5803	1.3286	0.5978	0.9498	0.5945	0.6371	0.5903	0.6279	0.5485
Slashdot	3.1169	3.2343	3.2422	3.2607	3.5241	3.1518	3.7520	4.3295	3.6161	2.9572
Social	3.3697	3.6543	3.7207	3.3983	3.5577	3.3760	3.5203	3.5200	3.5897	3.3527
Society	5.5787	5.5597	6.0830	5.6033	5.6930	5.5387	5.6443	5.6547	5.5753	5.5640
Yeast	6.5093	6.4842	6.7983	6.4635	6.8811	6.4831	6.5213	6.4482	6.5311	6.4035
AvgRank	3.93	5.20	8.80	4.80	8.03	3.80	6.57	6.27	6.20	1.40

TABLE V
EXPERIMENTAL RESULTS FOR THE COMPARISON OF AP ($\uparrow$).

	SRFS	MRDM	GRRO	CLEMLFS	SSFS	GLFS	LRDG	FMIOR	NMMFS	GLA-FMI
Arts	0.5491	0.5252	0.4764	0.5310	0.5137	0.5446	0.5271	0.5370	0.5214	0.5558
Business	0.8793	0.8770	0.8687	0.8753	0.8745	0.8797	0.8777	0.8775	0.8747	0.8817
Computers	0.6411	0.6383	0.6180	0.6408	0.6363	0.6453	0.6367	0.6437	0.6371	0.6449
Entertainment	0.5969	0.5951	0.5450	0.6053	0.5733	0.6067	0.5981	0.5940	0.5827	0.6211
Health	0.7393	0.6983	0.6385	0.7183	0.6803	0.7225	0.7155	0.7133	0.6988	0.7419
Image	0.7407	0.5835	0.7550	0.6078	0.7444	0.6001	0.7469	0.7442	0.7618	0.7839
Medical	0.7721	0.8247	0.4619	0.8384	0.7530	0.8184	0.7219	0.6378	0.7700	0.8500
PlantGo	0.8717	0.8736	0.8502	0.8741	0.8502	0.8912	0.7070	0.8014	0.8586	0.8928
Recreation	0.5363	0.5237	0.4194	0.5313	0.4803	0.5390	0.5203	0.5287	0.5020	0.5592
Reference	0.6595	0.6256	0.5870	0.6434	0.6278	0.6480	0.6380	0.6453	0.6435	0.6651
Scene	0.8273	0.8365	0.6520	0.8371	0.7458	0.8344	0.8340	0.8394	0.8233	0.8538
Slashdot	0.6176	0.5907	0.5934	0.5985	0.5601	0.6047	0.5165	0.4845	0.5514	0.6272
Social	0.7214	0.6926	0.6760	0.7185	0.6982	0.7365	0.7248	0.7218	0.7106	0.7347
Society	0.6031	0.6008	0.5554	0.5962	0.5901	0.6022	0.5958	0.5955	0.5975	0.6052
Yeast	0.7491	0.7482	0.7176	0.7477	0.7225	0.7500	0.7477	0.7572	0.7551	0.7592
AvgRank	3.93	6.27	9.03	4.90	8.23	3.13	6.37	5.60	6.40	1.13

TABLE VI
EXPERIMENTAL RESULTS FOR THE COMPARISON OF OE ($\downarrow$).

	SRFS	MRDM	GRRO	CLEMLFS	SSFS	GLFS	LRDG	FMIOR	NMMFS	GLA-FMI
Arts	0.5760	0.6003	0.6850	0.5950	0.6250	0.5763	0.6103	0.5827	0.6127	0.5547
Business	0.1187	0.1220	0.1327	0.1253	0.1233	0.1193	0.1240	0.1223	0.1253	0.1177
Computers	0.4393	0.4420	0.4587	0.4333	0.4337	0.4357	0.4417	0.4343	0.4403	0.4303
Entertainment	0.5373	0.5467	0.6227	0.5167	0.5730	0.5143	0.5330	0.5480	0.5660	0.5020
Health	0.3277	0.3890	0.4580	0.3507	0.4140	0.3543	0.3597	0.3597	0.3940	0.3190
Image	0.3970	0.5547	0.3790	0.5097	0.3880	0.5293	0.3860	0.3940	0.3580	0.3210
Medical	0.2853	0.2062	0.6806	0.1984	0.3085	0.2233	0.3628	0.4295	0.2992	0.1845
PlantGo	0.2077	0.1949	0.2462	0.2077	0.2333	0.1769	0.4154	0.2923	0.2282	0.1769
Recreation	0.5927	0.6103	0.7527	0.6010	0.6633	0.5887	0.6047	0.6010	0.6387	0.5600
Reference	0.4227	0.4680	0.5133	0.4453	0.4653	0.4383	0.4457	0.4420	0.4487	0.4250
Scene	0.2784	0.2709	0.5393	0.2659	0.3988	0.2751	0.2651	0.2625	0.2943	0.2400
Slashdot	0.4934	0.5272	0.5256	0.5108	0.5643	0.5087	0.6335	0.6510	0.5748	0.4870
Social	0.3650	0.3983	0.4287	0.3637	0.3967	0.3323	0.3520	0.3537	0.3737	0.3370
Society	0.4400	0.4393	0.4917	0.4487	0.4550	0.4407	0.4487	0.4520	0.4457	0.4373
Yeast	0.2432	0.2410	0.2486	0.2530	0.2541	0.2323	0.2410	0.2399	0.2356	0.2301
AvgRank	4.17	6.10	8.93	4.93	7.67	3.50	6.17	5.73	6.63	1.17

that for any two labels l' and l'' , if $h(x_i, l') > h(x_i, l'')$, then $\text{rank}(x_i, l') < \text{rank}(x_i, l'')$. The formal definitions of these metrics were described below.

$$\text{Cover} = \frac{1}{n_1} \sum_{i=1}^{n_1} \max_{l_d \in \mathcal{Y}_i} \text{rank}(x_i, l_d) - 1. \quad (35)$$

- Coverage (CV): measured the average number of positions required to include all relevant labels in the predicted ranking.
- Average Precision (AP): calculated the mean of the precision values obtained at the ranks of each relevant label.

TABLE VII
EXPERIMENTAL RESULTS FOR THE COMPARISON OF HL ($\downarrow$).

	SRFS	MRDM	GRRO	CLEMLFS	SSFS	GLFS	LRDG	FMIOR	NMMFS	GLA-FMI
Arts	0.0585	0.0601	0.0627	0.0583	0.0611	0.0584	0.0597	0.0598	0.0606	0.0580
Business	0.0270	0.0273	0.0283	0.0274	0.0280	0.0269	0.0273	0.0275	0.0279	0.0270
Computers	0.0396	0.0397	0.0411	0.0397	0.0398	0.0390	0.0398	0.0398	0.0402	0.0385
Entertainment	0.0594	0.0590	0.0633	0.0578	0.0613	0.0577	0.0585	0.0603	0.0606	0.0574
Health	0.0379	0.0429	0.0475	0.0406	0.0431	0.0397	0.0408	0.0410	0.0416	0.0376
Image	0.2016	0.0597	0.1922	0.0575	0.2010	0.0583	0.1912	0.2038	0.1892	0.1836
Medical	0.0164	0.0147	0.0275	0.0154	0.0185	0.0152	0.0220	0.0197	0.0180	0.0145
PlantGo	0.0417	0.0393	0.0481	0.0380	0.0487	0.0380	0.0652	0.0464	0.0489	0.0406
Recreation	0.0589	0.0593	0.0643	0.0599	0.0616	0.0583	0.0588	0.0590	0.0603	0.0577
Reference	0.0278	0.0318	0.0334	0.0286	0.0301	0.0288	0.0287	0.0289	0.0299	0.0275
Scene	0.1118	0.1049	0.1706	0.1031	0.1433	0.1080	0.1014	0.1054	0.1139	0.0945
Slashdot	0.0458	0.0462	0.0456	0.0457	0.0454	0.0445	0.0509	0.0512	0.0473	0.0443
Social	0.0245	0.0258	0.0272	0.0240	0.0259	0.0225	0.0239	0.0242	0.0245	0.0231
Society	0.0552	0.0557	0.0592	0.0549	0.0569	0.0555	0.0562	0.0566	0.0557	0.0543
Yeast	0.2020	0.2048	0.2286	0.2043	0.2216	0.2048	0.2054	0.1971	0.1999	0.2006
AvgRank	4.60	5.53	9.20	3.67	8.13	2.93	5.77	6.47	7.00	1.70

TABLE VIII
EXPERIMENTAL RESULTS FOR THE COMPARISON OF RL ($\downarrow$).

	SRFS	MRDM	GRRO	CLEMLFS	SSFS	GLFS	LRDG	FMIOR	NMMFS	GLA-FMI
Arts	0.1400	0.1445	0.1596	0.1437	0.1500	0.1418	0.1463	0.1453	0.1467	0.1397
Business	0.0396	0.0398	0.0436	0.0403	0.0427	0.0390	0.0404	0.0416	0.0408	0.0392
Computers	0.0883	0.0894	0.0970	0.0897	0.0931	0.0888	0.0888	0.0893	0.0927	0.0892
Entertainment	0.1157	0.1158	0.1272	0.1160	0.1219	0.1162	0.1150	0.1175	0.1189	0.1095
Health	0.0531	0.0614	0.0712	0.0582	0.0627	0.0561	0.0568	0.0597	0.0589	0.0522
Image	0.2248	0.1197	0.2073	0.1144	0.2130	0.1172	0.2182	0.2238	0.2078	0.1858
Medical	0.0563	0.0463	0.1151	0.0424	0.0559	0.0462	0.0654	0.0887	0.0573	0.0412
PlantGo	0.0528	0.0481	0.0521	0.0383	0.0513	0.0390	0.1385	0.0816	0.0507	0.0367
Recreation	0.1652	0.1741	0.2041	0.1667	0.1858	0.1682	0.1761	0.1736	0.1790	0.1609
Reference	0.0808	0.0870	0.0967	0.0832	0.0861	0.0834	0.0859	0.0825	0.0810	0.0783
Scene	0.1077	0.0955	0.2448	0.0992	0.1686	0.0991	0.1059	0.0975	0.1054	0.0890
Slashdot	0.1263	0.1324	0.1314	0.1331	0.1440	0.1281	0.1570	0.1820	0.1487	0.1198
Social	0.0629	0.0696	0.0714	0.0638	0.0677	0.0628	0.0657	0.0654	0.0679	0.0620
Society	0.1407	0.1404	0.1575	0.1425	0.1438	0.1401	0.1437	0.1435	0.1404	0.1403
Yeast	0.1806	0.1782	0.2044	0.1786	0.1996	0.1787	0.1802	0.1753	0.1786	0.1733
AvgRank	4.33	5.03	9.07	4.30	8.07	3.30	6.57	6.40	6.40	1.53

$$\text{AvgP} = \frac{1}{n_1} \sum_{i=1}^{n_1} \frac{1}{|\mathcal{Y}_i|} \sum_{l_d \in \mathcal{Y}_i} \frac{|\{l' \mid \text{rank}(x_i, l') \leq \text{rank}(x_i, l_d), l' \in \mathcal{Y}_i\}|}{\text{rank}(x_i, l_d)}. \quad (36)$$

- One Error (OE): computed the proportion of instances where the top-ranked label did not belong to the ground-truth label set.

$$\text{OneE} = \frac{1}{n_1} \sum_{i=1}^{n_1} \ll [\arg \max_{l_d \in \mathcal{L}} h(x_i, l_d)] \notin \mathcal{Y}_i \gg, \quad (37)$$

where $\ll \cdot \gg$ denoted the indicator function, which returned 1 if the condition held and 0 otherwise.

- Hamming Loss (HL): measured the fraction of misclassified labels over the total number of labels.

$$\text{HLoss} = \frac{1}{n_1} \sum_{i=1}^{n_1} \frac{1}{l} |\hat{\mathcal{Y}}_i \Delta \mathcal{Y}_i|, \quad (38)$$

where Δ represented the symmetric difference between two sets.

- Ranking Loss (RL): evaluated the proportion of label pairs in which an irrelevant label was ranked above a relevant one.

$$\text{RLoss} = \frac{1}{n_1} \sum_{i=1}^{n_1} \frac{1}{|\mathcal{Y}_i| |\bar{\mathcal{Y}}_i|} |\{(l', l'') \mid h(x_i, l') \leq h(x_i, l''), (l', l'') \in \mathcal{Y}_i \times \bar{\mathcal{Y}}_i\}|, \quad (39)$$

where $\bar{\mathcal{Y}}_i$ denoted the set of irrelevant labels for instance x_i .

3) *Compared Methods and Parameter Settings*: Nine MFS algorithms were considered for the comparative experimental analysis: SRFS [25], MRDM [27], GRRO [17], CLEMLFS [24], SSFS [29], GLFS [21], LRDG [28], FMIOR [4], and NMMFS [26]. For parameterized methods, SRFS optimized its parameters α , β , and γ via grid search within $\{0.01, 0.1, 0.3, 0.5, 0.7, 0.9, 1\}$. MRDM used a fixed α of 1, while β and γ were optimized by grid search within $\{0.01, 0.1, \dots, 100\}$. Similarly, α_1 and β in GRRO, α , β and γ in SSFS and LRDG, α and β in FMIOR, and α , β , γ and η in NMMFS were optimized by a grid search within $\{0.001, 0.01, \dots, 1000\}$. For CLEMLFS, λ_1 and λ_2 were optimized by a grid search within $\{0.01, 0.1, \dots, 100\}$. For GLFS, α was fixed at 1, while β and γ were optimized by

a grid search within $\{0.001, 0.01, \dots, 1000\}$. For the proposed GLA-FMI method, the parameters α , β , and γ were optimized through grid search within the range $\{0.001, 0.01, \dots, 1000\}$. Additionally, the parameters r_1 and r_2 were both set to the same value r , which was optimized through grid search within the range $\{0.5, 1.0, 1.2, 1.4, 1.6, 1.8, 2.0\}$. Meanwhile, to ensure experimental fairness, the number of selected features was determined by $\text{round}(\tau \times m)$, where $\tau \in \{0.4, 0.3, 0.2, 0.1\}$ was chosen based on the feature dimensionality m [18].

B. Analysis of Experimental Results

Tables IV-VIII reported the performance of various algorithms across fifteen datasets, evaluated using five distinct evaluation metrics. For each dataset, the top-performing result was marked in bold. Additionally, “AvgRank” denoted the average rank of each algorithm under a specific metric, with lower values reflecting better overall performance. From the results, it was observed that the GLA-FMI method showed consistently strong performance across all five evaluation metrics. Specifically, it ranked first on 73.33%, 86.67%, 86.67%, 66.67%, and 73.33% of the datasets for the CV, AP, OE, HL, and RL metrics, respectively. Moreover, GLA-FMI achieved the lowest average rank in each metric, highlighting its overall advantage compared to the other methods.

To highlight the advantages of GLA-FMI, feature addition experiments were carried out to assess performance trends across five evaluation metrics. Figures 24 presented the results on the *Entertainment*, *Health*, and *Medical* datasets. The horizontal axis represented the number of selected features, while the vertical axis showed the corresponding values of the evaluation metrics. In general, as the number of selected features increased, the performance of all methods initially improved, leveled off, and then potentially declined. As shown in the figure, when the number of selected features was small (especially below 50), SRFS performed better than GLA-FMI, possibly because its sparse supplementation norm more effectively focused on a few highly discriminative features in the early stage. As the number of features increased, the global feature importance guidance and redundancy-aware constraints introduced by GLA-FMI gradually took effect, leading to more stable performance improvements and eventually surpassing SRFS in the later stages. Compared with other methods except SRFS, GLA-FMI consistently maintained a leading position once the number of selected features exceeded 10. Overall, GLA-FMI demonstrated stronger stability and adaptability throughout the feature selection process, reflecting superior overall performance.

C. Statistical Analysis

To further validate the superiority of the proposed algorithm, both the Friedman test [47] and the Bonferroni-Dunn test [48] were conducted. The Friedman test was initially applied to determine whether there were significant differences between the algorithms. The results, presented in Table IX, indicated that, at a significance level of $\psi = 0.05$, the Friedman statistic F_F for each metric exceeded the critical value, confirming statistically significant differences between GLA-FMI and the

other algorithms. To investigate these differences in performance further, the Bonferroni-Dunn test was carried out with GLA-FMI as the benchmark. The critical distance (CD) was calculated using the following formula:

$$CD = q_\psi \sqrt{\frac{a_k(a_k + 1)}{6N}}, \quad (40)$$

where $q_\psi = 2.773$ for a confidence level of $\psi = 0.05$. Given the number of algorithms $a_k = 10$ and datasets $N = 15$, the CD was computed as 3.0657.

TABLE IX
FRIEDMAN TEST RESULTS ON FIVE EVALUATION METRICS ($a_k=10$, $N=15$).

Evaluation metrics	F_F	Critical value ($\psi = 0.05$)
CV	14.7054	1.955
AP	21.0131	
OE	15.5814	
HL	25.5554	
RL	17.4246	

Figure 5 presented the CD diagrams for five evaluation metrics, where algorithms with an average ranking difference smaller than the computed CD value were connected to GLA-FMI by a red line. The results showed that GLA-FMI outperformed GRRO, SSFS, LRDG, MRDM, FMIO, and N-MMFS across all evaluation metrics. Additionally, GLA-FMI performed better than CLEMLFS on all metrics except HL and RL. Although the differences between GLA-FMI and both SRFS and GLFS were not statistically significant on any metric, GLA-FMI consistently ranked higher than both. One possible reason was that SRFS effectively retained discriminative features and suppressed redundancy to some extent through the sparse supplementation regularization, while GLFS enhanced the specificity of feature selection and model expressiveness by jointly learning shared features among similar labels and label-specific features. However, both SRFS and GLFS failed to sufficiently model global feature importance and could not effectively distinguish between redundant features, often resulting in the retention of multiple highly redundant features. In contrast, GLA-FMI incorporated the FMI framework for global feature importance evaluation and introduced an explicit redundancy-aware mechanism, which improved the discriminability of the selected feature subset while effectively controlling redundancy, thereby demonstrating superior overall performance and robustness.

D. Parameter Sensitivity Analysis

In this section, the impact of four parameters (α , β , γ , and r) on the classification performance of GLA-FMI was analyzed. Specifically, α controlled the penalty strength for feature redundancy, β regulated the degree of suppression for the weight rows of unactivated features, γ adjusted the influence of global feature importance, and r represented the exponent parameter in the penalty term. To investigate the sensitivity of each parameter during the feature selection process, the target parameter was varied within its predefined search space while keeping the other parameters fixed at their optimal values.

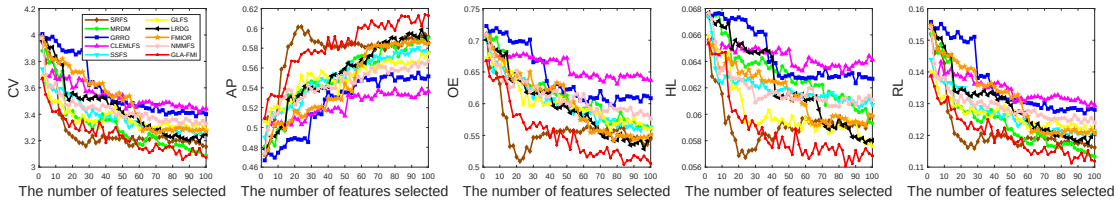


Fig. 2. Algorithm performance on the *Entertainment* dataset with varying numbers of selected features.

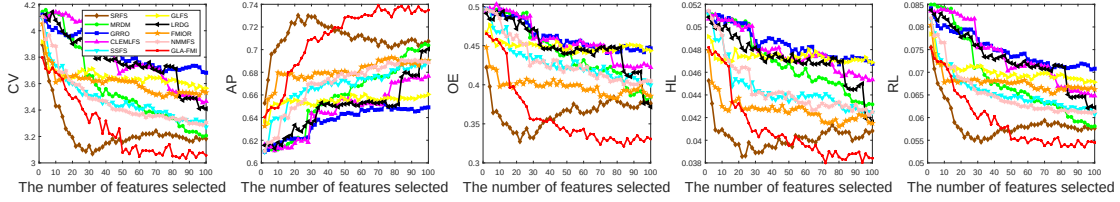


Fig. 3. Algorithm performance on the *Health* dataset with varying numbers of selected features.

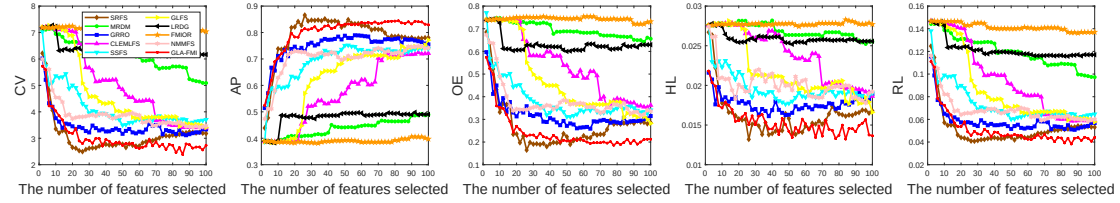


Fig. 4. Algorithm performance on the *Medical* dataset with varying numbers of selected features.

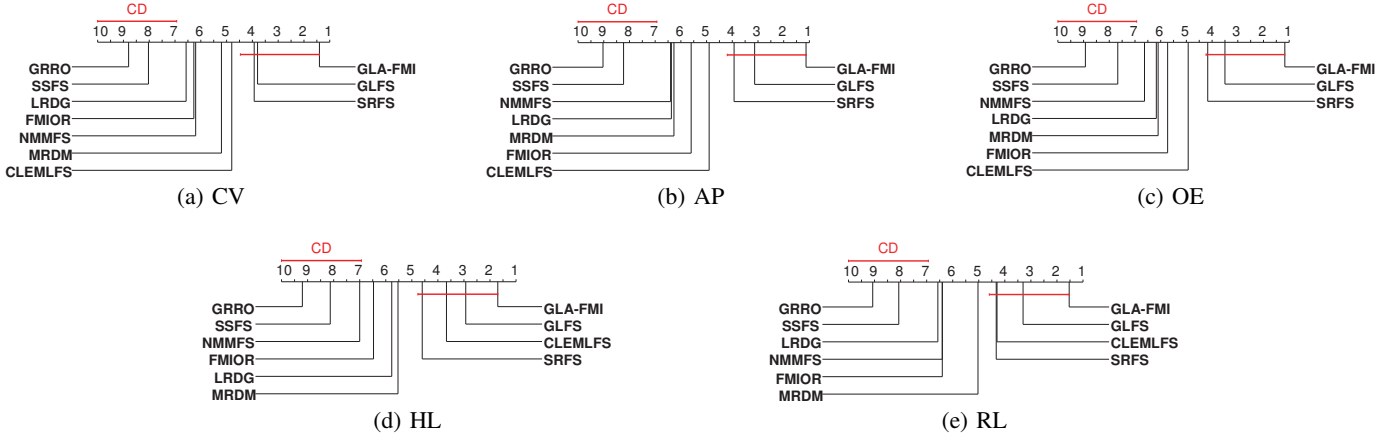


Fig. 5. Bonferroni-Dunn test results for five evaluation metrics.

Figures 6 and 7 showed the experimental results for two evaluation metrics (CV and AP) on the *Arts* dataset. The results indicated that for the *Arts* dataset, when α was less than 0.1, it effectively balanced the feature redundancy penalty, but higher values suppressed the selection of useful features. The value of β should be small (ranging from 10^{-3} to 10), as larger values excessively suppressed the weights of unactivated features, negatively affecting model performance. When γ was larger (e.g., 100), the model performed best, highlighting the importance of global feature importance in guiding the model to select features with higher discriminative power. Finally, the value of r should remain small (between 0.5 and 1.2) to keep the penalty term approximately linear, as larger values could lead to extreme solutions during optimization.

E. Ablation Studies

To verify the effectiveness of the core components based on the three types of relationship modeling in GLA-FMI, ablation studies were conducted by individually removing each module and evaluating its impact on performance. GLA-FMI1 removed the label importance estimation module, thereby ignoring label correlations and treating all labels as equally important by setting $\theta = \frac{1}{l}\mathbf{1}^{l \times 1}$. GLA-FMI2 removed the redundancy constraint term, neglecting feature redundancy by setting $\alpha = 0$. GLA-FMI3 removed the global feature importance guidance term, thereby ignoring feature-label relevance by setting $\gamma = 0$. The ablation results on the *Arts*, *Business*, *Computers*, and *Entertainment* datasets were presented in

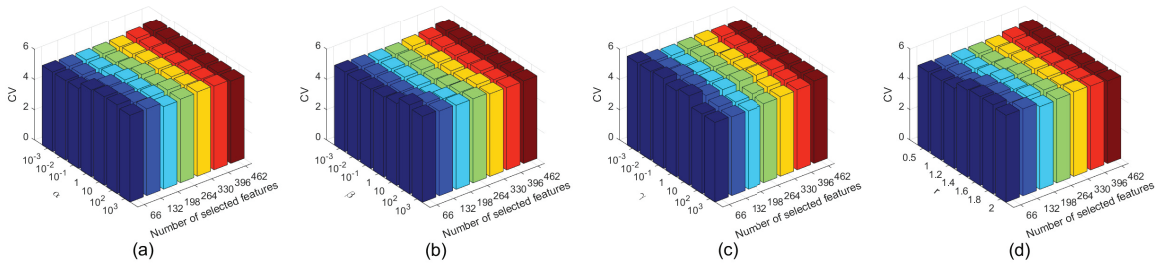


Fig. 6. Parameter sensitivity analysis for GLA-FMI on the *Art* dataset using CV. (a) α . (b) β . (c) γ . (d) r .

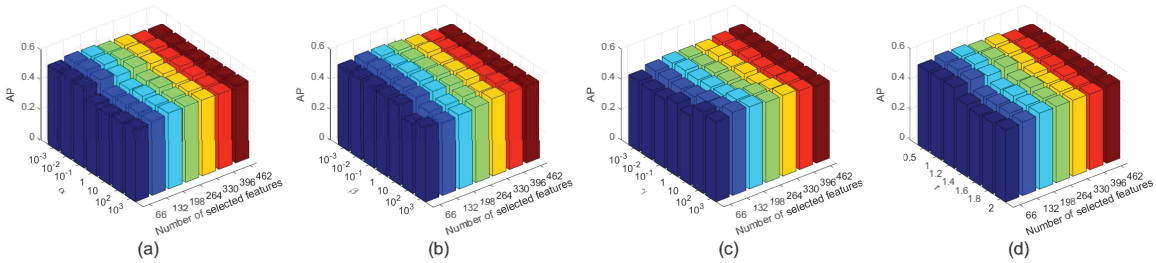


Fig. 7. Parameter sensitivity analysis for GLA-FMI on the *Art* dataset using AP. (a) α . (b) β . (c) γ . (d) r .

Figure 8. As shown in the figure, GLA-FMI3 exhibited the most significant performance degradation, indicating that the global feature importance guidance term played a crucial role in enhancing model performance. This was likely because the term guided the model to focus on more representative feature subsets, which improved prediction accuracy. Without this component, the models discriminative ability was considerably weakened. In comparison, GLA-FMI1 and GLA-FMI2 showed relatively smaller performance declines, suggesting that although the label importance estimation module and the redundancy constraint term contributed to performance improvement, their influence was less critical than that of the global feature importance guidance term. Overall, the complete GLA-FMI model consistently outperformed its ablated variants across the four datasets, validating the effectiveness of the three core components and demonstrating the necessity of modeling all three types of relationships.

F. Convergence Analysis and Running Time Analysis

In Section IV-F, the theoretical convergence of the GLA-FMI algorithm was rigorously established. To further validate its empirical convergence performance, experiments were conducted on several benchmark datasets, with results reported for six representative ones. As illustrated in Figure 9, the objective function value dropped rapidly and stabilizes within approximately 5 iterations across different datasets, demonstrating fast and stable convergence behavior. These findings indicated that GLA-FMI not only had a solid theoretical foundation but also exhibited high computational efficiency and practical feasibility.

In addition, the running time of each algorithm under optimal parameter settings was compared. As shown in Table X, for GLA-FMI, GRRO, and FMIOR, the relationship matrices were precomputed and treated as prior knowledge, since their calculation was independent of the model opti-

mization process. Therefore, the reported time reflected only the computational cost of the models themselves. Overall, SRFS, GRRO, SSFS, FMIOR, and NMMFS had running times within a similar order of magnitude. NMMFS was slightly slower, possibly due to the use of a small step size in the gradient descent method, which slowed down convergence. MRDM showed lower efficiency because it relied on feature decomposition operations with a computational complexity of $O(n^3)$. CLEMLFS required more time due to the high cost of computing instance correlations. GLFS incurred significant iteration overhead as it involved solving Lyapunov equations and quadratic programming problems. GLA-FMI performed efficiently on datasets with low-dimensional features, but its runtime increased noticeably on datasets with high-dimensional features due to its $O(m^3)$ complexity. In summary, although GLA-FMI achieved strong performance, there remained room for improvement in computational efficiency when dealing with high-dimensional features.

VI. CONCLUSION

This paper proposes a global-guided label-aware adaptive MFS method based on fuzzy mutual information (GLA-FMI). To address several key limitations of existing embedded MFS methods, GLA-FMI designs a convergent recursive propagation algorithm to reliably estimate label importance, proposes an adaptive feature activation strategy for explicit feature selection, and employs fuzzy mutual information to capture global feature importance from raw data. Additionally, a redundancy-aware constraint and a feature sparsity regularization term are incorporated to improve the model's robustness and interpretability. Experiments on multiple benchmark multi-label datasets demonstrate that GLA-FMI significantly outperforms existing state-of-the-art methods. Future work will explore the extension of this method to partial multi-label feature selection, where the label space contains false positive

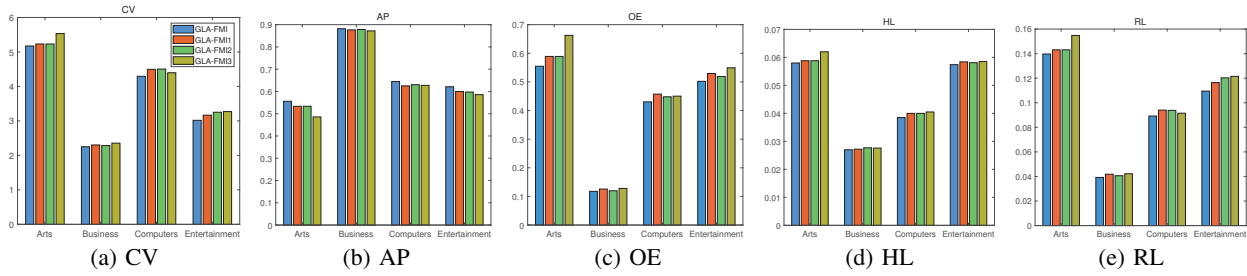


Fig. 8. (a)(e) show the performance comparison of four model variants across five evaluation metrics.

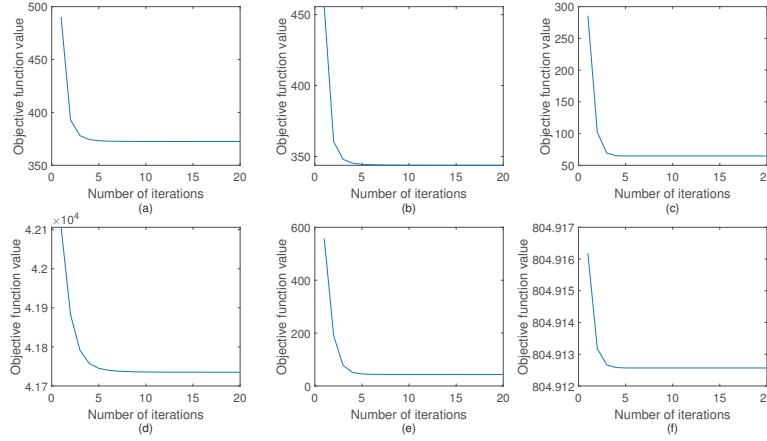


Fig. 9. Convergence of objective function values. (a) *Entertainment*. (b) *Health*. (c) *Medical*. (d) *Reference*. (e) *Social*. (f) *Yeast*.

TABLE X
RUNNING TIME (IN SECOND).

	SRFS	MRDM	GRRO	CLEMLFS	SSFS	GLFS	LRDG	FMIOR	NMMFS	GLA-FMI
Arts	4.70	83.49	4.46	37.08	5.49	6.65	6.64	5.58	9.65	7.50
Business	4.92	84.22	4.35	31.86	5.08	9.96	6.39	5.48	8.04	8.59
Computers	4.86	84.03	4.55	32.16	5.14	40.32	7.09	5.41	14.50	21.29
Entertainment	4.39	83.02	4.18	32.84	4.80	8.97	6.35	5.84	12.89	22.54
Health	4.66	83.08	4.21	36.35	4.98	8.78	6.41	5.28	12.25	28.07
Image	0.69	12.81	0.63	23.28	0.71	2.29	1.08	1.09	1.33	0.88
Medical	1.30	3.20	0.37	1.22	1.20	4.15	0.70	1.03	3.44	254.54
PlantGo	4.43	15.45	0.80	11.68	6.06	13.09	2.40	2.27	14.83	541.79
Recreation	4.24	83.39	4.10	37.56	4.60	16.70	6.22	5.08	11.76	9.58
Reference	5.46	84.98	4.93	40.59	5.68	11.52	7.49	5.94	18.93	43.97
Scene	0.95	18.83	0.89	16.66	0.95	6.37	1.47	1.62	1.76	1.27
Slashdot	3.26	69.36	2.86	44.51	4.27	246.40	5.85	3.85	26.36	47.45
Social	4.83	86.13	3.99	42.51	4.72	276.23	7.72	4.92	29.46	203.68
Society	4.52	84.24	4.20	40.83	4.95	7.74	6.51	5.24	13.19	23.13
Yeast	0.83	33.52	0.78	68.69	1.07	3.57	1.45	1.28	1.60	0.87

labels.

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